

Answer 1

Since X is bounded, so $\mathbb{E}[X]$ and $\mathbb{E}[X^2]$ are both finite and the variance makes sense. So, $\mu = \mathbb{E}[X]$. Since $0 \leq X \leq c$, taking expectations gives $0 \leq \mu \leq c$.

Proving the hint. Whenever $X \in [0, c]$ (which happens with probability one), both X and $c - X$ are nonnegative, so their product is too:

$$X(c - X) \geq 0 \quad \text{almost surely.}$$

A random variable that is nonnegative almost surely has a nonnegative expectation. Expanding and using linearity,

$$0 \leq \mathbb{E}[X(c - X)] = c\mathbb{E}[X] - \mathbb{E}[X^2],$$

which rearranges to

$$\mathbb{E}[X^2] \leq c\mathbb{E}[X]. \tag{H}$$

Using it to bound the variance. Plugging (H) into $\text{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2$ gives

$$\text{Var}(X) \leq c\mu - \mu^2 = \mu(c - \mu).$$

So now it's enough to show that $\mu(c - \mu)$ can never be bigger than $c^2/4$. Completing the square,

$$\mu(c - \mu) = \frac{c^2}{4} - \left(\mu - \frac{c}{2}\right)^2 \leq \frac{c^2}{4},$$

because a square is never negative. Putting the two inequalities together,

$$\text{Var}(X) \leq \mu(c - \mu) \leq \frac{c^2}{4},$$

which is exactly inequality (1). □

This bound tight as if X is 0 or c with probability $\frac{1}{2}$ each, then $\mathbb{E}[X] = c/2$ and $\mathbb{E}[X^2] = c^2/2$, so $\text{Var}(X) = c^2/2 - c^2/4 = c^2/4$. This is actually the only case where equality holds: we need $X(c - X) = 0$ almost surely (so X only takes the values 0 and c) and we need $\mu = c/2$.

Exercise 2

First, $f(a)$ is finite for every a : since $\mathbb{E}[X^2] = \sigma^2 + \mu^2 < \infty$ and $(X - a)^2 = X^2 - 2aX + a^2$, each piece has a finite expectation.

Adding and subtracting μ inside the square. Writing $X - a = (X - \mu) + (\mu - a)$ and expanding,

$$f(a) = \mathbb{E}[(X - \mu)^2] + 2(\mu - a)\mathbb{E}[X - \mu] + (\mu - a)^2.$$

Here I pulled $(\mu - a)$ out of the expectation since it's just a number. The first term is σ^2 by the definition of variance, and the middle term vanishes because $\mathbb{E}[X - \mu] = \mu - \mu = 0$. So

$$f(a) = \sigma^2 + (a - \mu)^2.$$

Written this way, the answer is easy to read off. The term σ^2 doesn't depend on a , and $(a - \mu)^2 \geq 0$ with equality only when $a = \mu$. So $f(a) \geq \sigma^2 = f(\mu)$ for every real a , and the minimum is reached at exactly one point:

$$\boxed{a^* = \mu = \mathbb{E}[X]} \quad \text{and} \quad \boxed{f(a^*) = \sigma^2 = \text{Var}(X)}.$$

As a quick sanity check with calculus: $f(a) = \mathbb{E}[X^2] - 2a\mu + a^2$ is a parabola in a with $f'(a) = 2(a - \mu)$ and $f''(a) = 2 > 0$. The only critical point is $a = \mu$, and since f is convex it's the global minimum. Plugging in gives $f(\mu) = \mathbb{E}[X^2] - \mu^2 = \sigma^2$, which agrees.

In words: the mean is the best constant guess for X in the mean-squared-error sense, and the error you're left with is exactly the variance.

Exercise 3

The whole exercise rests on the geometric series. For $|x| < 1$,

$$\sum_{k=0}^{\infty} x^k = \frac{1}{1-x}.$$

A power series can be differentiated term by term inside its radius of convergence (here, $|x| < 1$). Differentiating once and then twice gives

$$\sum_{k=1}^{\infty} k x^{k-1} = \frac{1}{(1-x)^2}, \tag{G1}$$

$$\sum_{k=2}^{\infty} k(k-1) x^{k-2} = \frac{2}{(1-x)^3}. \tag{G2}$$

We'll use these with $x = q$, which is fine because $0 < q < 1$. It also helps to remember that $1 - q = p$.

Part 1: $\mathbb{E}[X] = 1/p$

By definition, and then using (G1) with $x = q$,

$$\mathbb{E}[X] = \sum_{k=1}^{\infty} k q^{k-1} p = p \sum_{k=1}^{\infty} k q^{k-1} = p \cdot \frac{1}{(1-q)^2} = \frac{p}{p^2} = \frac{1}{p}.$$

(All the terms are nonnegative, so there's no issue with the series being well defined.) This matches intuition: if each trial succeeds with probability p , you expect to wait about $1/p$ trials.

Part 2: $\mathbb{E}[X(X-1)]$ and the variance

The $k = 1$ term is zero because $k(k-1) = 0$ there, so the sum really starts at $k = 2$. Pulling out a factor of pq so that the powers line up with (G2),

$$\mathbb{E}[X(X-1)] = \sum_{k=2}^{\infty} k(k-1) q^{k-1} p = pq \sum_{k=2}^{\infty} k(k-1) q^{k-2} = pq \cdot \frac{2}{(1-q)^3} = \frac{2pq}{p^3}.$$

So

$$\mathbb{E}[X(X-1)] = \frac{2q}{p^2} = \frac{2(1-p)}{p^2}.$$

To get the variance, notice that $X^2 = X(X-1) + X$. Taking expectations,

$$\mathbb{E}[X^2] = \frac{2q}{p^2} + \frac{1}{p},$$

and therefore

$$\text{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2 = \frac{2q}{p^2} + \frac{1}{p} - \frac{1}{p^2} = \frac{2q + p - 1}{p^2}.$$

Since $p - 1 = -q$, the numerator simplifies to $2q - q = q$, giving

$$\text{Var}(X) = \frac{q}{p^2} = \frac{1-p}{p^2}.$$

We compute $\mathbb{E}[X(X-1)]$ rather than $\mathbb{E}[X^2]$ directly because the factor $k(k-1)$ is exactly what the second derivative of the geometric series produces.

Part 3: $\mathbb{E}[1/X] \geq p$ without computing $\mathbb{E}[1/X]$

First, $\mathbb{E}[1/X]$ is finite: $X \geq 1$, so $0 < 1/X \leq 1$.

Now use Jensen's inequality with $\varphi(x) = 1/x$. On $(0, \infty)$ we have $\varphi''(x) = 2/x^3 > 0$, so φ is convex there. The variable X takes values in $(0, \infty)$ and has finite mean ($1/p$, from Part 1), so Jensen applies and gives

$$\mathbb{E}\left[\frac{1}{X}\right] = \mathbb{E}[\varphi(X)] \geq \varphi(\mathbb{E}[X]) = \frac{1}{\mathbb{E}[X]} = \frac{1}{1/p} = p.$$

That's the claim, $\mathbb{E}[1/X] \geq p$.

Another way to see the same thing is Cauchy-Schwarz:

$$1 = \left(\mathbb{E}\left[\sqrt{X} \cdot \frac{1}{\sqrt{X}}\right] \right)^2 \leq \mathbb{E}[X] \mathbb{E}\left[\frac{1}{X}\right] = \frac{1}{p} \mathbb{E}\left[\frac{1}{X}\right],$$

which again gives $\mathbb{E}[1/X] \geq p$.

The inequality is actually strict. $1/x$ is strictly convex, and X isn't constant (for example, $X = 1$ and $X = 2$ both have positive probability when $0 < p < 1$).

Exercise 4

Throughout, let $R_1, \dots, R_n$ be the outcomes of the rolls. They are independent and each is uniform on $\{1, \dots, 6\}$.

Part 1: $\mathbb{E}[I_i]$, $\mathbb{E}[X]$, and the case $n = 1$

The expectation of an indicator is just the probability of its event: $\mathbb{E}[I_i] = 1 \cdot \mathbb{P}(A_i) + 0 \cdot \mathbb{P}(A_i^c) = \mathbb{P}(A_i)$.

Face i never appears exactly when every single roll misses it:

$$A_i = \{R_1 \neq i\} \cap \{R_2 \neq i\} \cap \cdots \cap \{R_n \neq i\}.$$

Each roll misses face i with probability $5/6$, and the rolls are independent, so we can multiply:

$$\mathbb{E}[I_i] = \mathbb{P}(A_i) = \left(\frac{5}{6}\right)^n.$$

This is the same for every $i = 1, \dots, 6$.

For X , linearity of expectation does the rest. Linearity holds even though (as Part 2 shows) the I_i are not independent:

$$\mathbb{E}[X] = \sum_{i=1}^6 \mathbb{E}[I_i] = 6 \left(\frac{5}{6}\right)^n.$$

When $n = 1$ this gives $\mathbb{E}[X] = 6 \cdot \frac{5}{6} = 5$. That makes sense: with one roll, exactly one face shows up and the other five don't, so $X = 5$ no matter what happens.

Part 2: $\mathbb{E}[I_i I_j]$ for $i \neq j$, and why I_i, I_j aren't independent

The product of two indicators is the indicator of the intersection, so $I_i I_j = \mathbf{1}_{A_i \cap A_j}$ and $\mathbb{E}[I_i I_j] = \mathbb{P}(A_i \cap A_j)$.

The event $A_i \cap A_j$ means neither face i nor face j ever appears, i.e. every roll lands on one of the other four faces. Each roll does this with probability $4/6 = 2/3$, and again the rolls are independent, so

$$\mathbb{E}[I_i I_j] = \left(\frac{4}{6}\right)^n = \left(\frac{2}{3}\right)^n.$$

If I_i and I_j were independent, we would have $\mathbb{E}[I_i I_j] = \mathbb{E}[I_i] \mathbb{E}[I_j]$. Let's compare the two sides:

$$\mathbb{E}[I_i] \mathbb{E}[I_j] = \left(\frac{5}{6}\right)^{2n} = \left(\frac{25}{36}\right)^n, \quad \mathbb{E}[I_i I_j] = \left(\frac{2}{3}\right)^n = \left(\frac{24}{36}\right)^n.$$

Since $24/36 < 25/36$ and $n \geq 1$, the second is strictly smaller than the first. So $\mathbb{E}[I_i I_j] \neq \mathbb{E}[I_i] \mathbb{E}[I_j]$, and I_i and I_j are **not independent**.

Intuitively, the dependence goes in the negative direction. If you learn that face i never showed up, all n rolls landed on the other five faces, so face j had a better chance per roll to appear. Concretely,

$$\mathbb{P}(A_j \mid A_i) = \frac{(4/6)^n}{(5/6)^n} = \left(\frac{4}{5}\right)^n < \left(\frac{5}{6}\right)^n = \mathbb{P}(A_j),$$

so $\text{Cov}(I_i, I_j) = (2/3)^n - (25/36)^n < 0$.